

Welcome To Our Presentation

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What is CapScan? & Purpose

What is CapScan?

CapScan is a desktop application that helps users verify the authenticity of memory cards and USB drives. It checks the actual storage capacity, detects fake, copied, or corrupted devices, and provides clear, easy-to-read results.

Purpose:

- Ensure devices are genuine and reliable
- Prevent data loss from fake or faulty drives
- Provide accurate capacity and speed information

Why Use Database?

A database is used to store models of memory cards from various companies. This allows CapScan to verify the brand, model, and authenticity of connected drives and maintain a record of previously tested devices, helping users track and identify fake or copied cards efficiently.



Why CapScan is Needed and What It Offers ?

Why CapScan is Needed:

- **Many memory cards & USB drives misreport capacity.**
- **Fake or low-quality devices cause data loss or corrupted files.**
- **Reliable tools for testing storage are hard to find.**

What CapScan Offers:

- **Lightweight and user-friendly interface**
- **Verifies authenticity** of storage devices
- **Checks real capacity** vs. advertised
- **Measures performance** to prevent data loss

Key Features

- 1 Confirm device authenticity instantly
- 2 Validate real vs. advertised storage
- 3 Detect corrupted or fake sectors
- 4 Benchmark read/write speeds
- 5 Summarize usable capacity and errors
- 6 Export reports in TXT or CSV
- 7 Toggle between multiple languages

Tools & Outcome:

Tools Used:

- Python programming
- Tkinter GUI framework
- MySQL / SQLite for SD card model verification
- File I/O and OS-level benchmarking
- Visual Studio Code IDE

Flaws Detected:

- Fake or copied storage devices
- Misreported capacity or slow performance
- Corrupted or unreliable sectors
- Prevents data loss for users

Conclusion & Future Enhancements:

Conclusion:

CapScan is a reliable and user-friendly tool that allows users to verify the authenticity, capacity, and performance of memory cards and USB drives. By detecting fake, copied, or slow devices, it helps prevent data loss and ensures that storage devices are genuine and trustworthy.

Future Enhancements:

- Cloud-based report syncing for test history
- Mobile platform support for on-the-go verification
- Advanced SD card model verification for more brands
- Graphical dashboards for analysis and trends

